

# ECON 311

## Lecture 4: Economic Growth

# Introduction to Long-Run Growth

## A Production Model

# Introduction to Growth

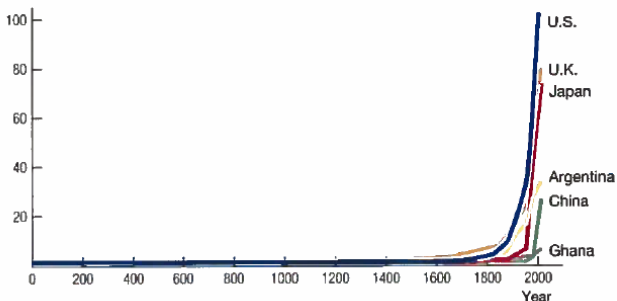
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- Standards of living in industrialized countries have greatly increased over the past few centuries.
  - ▶ E.g. Averages incomes today in North America and Western Europe are between 50 and 300 times larger than two centuries ago.
- This increase in standards of living has been a very recent phenomena in human history.
  - ▶ For most of their history (before the Industrial Revolution), humans lived barely above subsistence levels.
- Economic growth has been very uneven across time.
  - ▶ E.g. the productivity growth slowdown of 1990's
- Economic growth has been very uneven across countries.
  - ▶ E.g. growth miracles (Japan, South Korea, Singapore), growth disasters (Argentina, Sub-Saharan countries)

# Economic Growth Over the Very Long Run

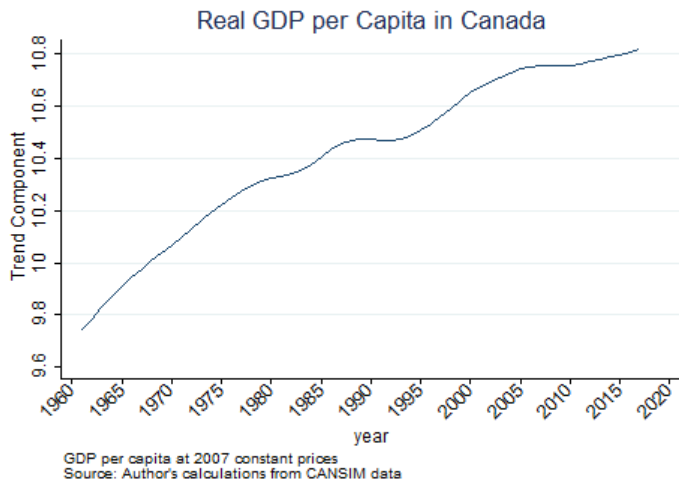
## Economic Growth over the Very Long Run in Six Countries

Per capita GDP  
(multiple of 300 dollars)



Source: Macroeconomics, Jones

## Economic Growth Over Time



# Defining Economic Growth

- Let  $g$  be a constant growth rate over a period of time, then ...

$$y_{t+T} = y_t(1 + g)^T$$

- Important properties of growth rates:

- (Rule of 70) Let  $y_{t+T} = 2y_t$  and , then

$$T \approx \frac{0.7}{g}$$

- In general, the time it takes for a series to increase by a certain ratio does not depend on the initial level.
- Algebraic properties of growth rates (let the variable  $i$  grow at a constant growth rate  $g_i$ ):

$$z = \frac{x}{y}$$

$$\Rightarrow g_z = g_x - g_y$$

$$z = x \cdot y$$

$$\Rightarrow g_z = g_x + g_y$$

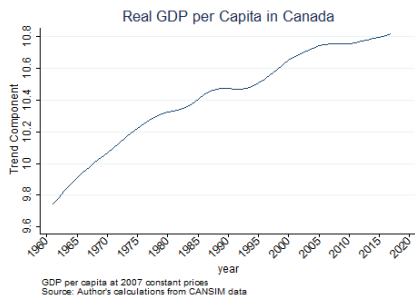
$$z = x^a \text{ (where } a \text{ is a constant)} \Rightarrow g_z = a \cdot g_x$$

## GDP per capita trend revisited

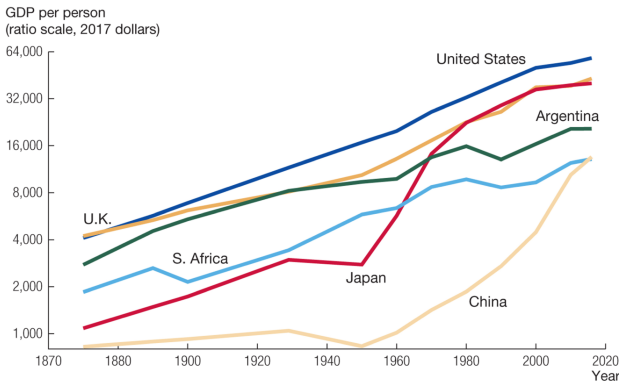
- We can calculate average growth rates over a period of time  $T$  as follows:

$$\bar{g} = \left( \frac{y_{t+T}}{y_t} \right)^{\frac{1}{T}} - 1$$

- To separate the trend from the cycle, we choose a particular period of time and change the annual GDP per capita by that rate



# Graphing Growth Rates



Source: The Maddison-Project, [www.ggd.net/maddison/](http://www.ggd.net/maddison/).  
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Source: Macroeconomics, Jones



# The power of economic growth

| Country       | Period    | Real GDP per Person at Beginning of Period* | Real GDP per Person at End of Period* | Growth Rate (per year) |
|---------------|-----------|---------------------------------------------|---------------------------------------|------------------------|
| United States | 1870–2010 | 4 166                                       | 48 635                                | 1.77                   |
| Indonesia     | 1900–2010 | 926                                         | 4 306                                 | 1.41                   |
| Pakistan      | 1900–2010 | 766                                         | 2 843                                 | 1.20                   |
| Bangladesh    | 1900–2010 | 648                                         | 1 854                                 | 0.96                   |

Real GDP is measured in 2010 Canadian dollars.

Source: Mankiw, N., Kneebone, R., & McKenzie, K. (2017). *Principles of Macroeconomics* (7th Canadian ed.). Toronto, ON: Nelson Education

- At average growth rates it will take Bangladesh around 340 years, Pakistan around 237 years and Indonesia around 172 years to reach levels of real GDP per capita similar to US in 2010.

# The power of economic growth

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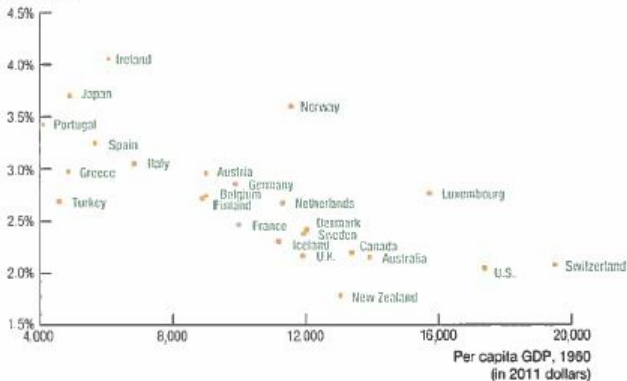
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- If average growth were to increase by a *quarter* of a percent, it would take Bangladesh around 270 years (diff. of 70 years), Pakistan around 196 years (diff. of 41 years) and Indonesia around 146 years (diff. of 26 years) to reach levels of real GDP per capita similar to US in 2010.

# Growth Rates Among Developed Countries

## Growth Rates in the OECD, 1960–2014

Per capita GDP growth  
(1960–2014)



Source: Advanced Macroeconomics, D. Romer

# Long-Run Growth

- Given the importance of long-run growth for well-being, we will study the classical model of economic growth.
- The goal is to learn the mechanics of the model and its insights concerning the dynamics of long-run growth across time and countries.
- The ultimate objective of research on economic growth is to study possible economic policies for raising growth in poor countries to bring GDP per capita closer to the level of rich countries.
  - ▶ To that end, we will confront this model with the data to see if it offers meaningful answers to our central objective.

## Introduction to Long-Run Growth

### A Production Model

# Introduction

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- To understand the causes of the great differences in production across time and countries, we first write down the following production function:

$$Y = F(K, L) = \bar{A}K^{\alpha}L^{(1-\alpha)} \quad , \quad \alpha \leq 1$$

- This production function has several theoretical properties that facilitates the analysis of empirical implications of differences in  $K$  and  $L$ .

# Properties of the Production Function

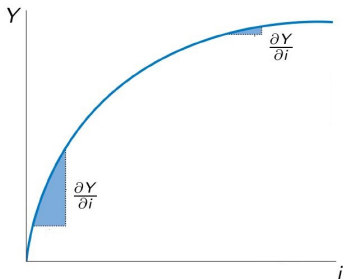
The Cobb-Douglas Production Function has the following properties:

1. Diminishing returns to  $K$  and  $L$ :

Diminishing Returns to  $i$  for  $i = K, L$

$$MPK = \frac{\partial Y}{\partial K} = \alpha \bar{A} \left( \frac{L}{K} \right)^{(1-\alpha)}$$

$$MPL = \frac{\partial Y}{\partial L} = (1 - \alpha) \bar{A} \left( \frac{K}{L} \right)^{\alpha}$$



- Differences in  $K$  accumulation per person has different implications at different levels of development

# Properties of the Production Function

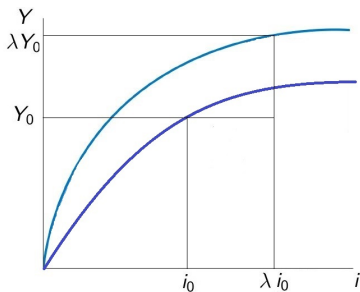
The Cobb-Douglas Production Function has the following properties:

## 2. Constant Returns to Scale:

$$F(\lambda K, \lambda L) = \lambda F(K, L)$$

$$F(\lambda K, \lambda L) = \lambda \underbrace{\bar{A} K^{\alpha} L^{(1-\alpha)}}_{F(K,L)}$$

Constant Returns to scale



- Population size matters for total income but does not for income per capita



# Solving for Production Parameters

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- To take this model to the data, we need ...
  - ▶ to assign value to the parameter,  $\alpha$
  - ▶ and study how differences in the exogenous variables,  $K$ ,  $L$ ,  $\bar{A}$ , result in differences in endogenous variable,  $Y$ .
- It turns out that from the optimal decision of firms to rent  $K$  and hire  $L$ , we obtain:

$$\frac{wL^*}{Y^*} = (1 - \alpha) \quad , \quad \frac{rK^*}{Y^*} = \alpha$$

- We can obtain this from the income approach to measuring GDP:  
 $\alpha = \frac{1}{3}$

# Development Accounting

- Our main concern is to measure GDP per capita differences, so we expressed the production in per capita terms:

$$y = \bar{A}k^{\frac{1}{3}}, \quad y = \frac{Y}{L}, \quad k = \frac{K}{L}$$

- We are going to assume that a country uses all of its resources to produce:

$$k = \bar{k}, \quad \bar{k} = \frac{\bar{K}}{\bar{L}}$$

- We observe differences in  $\bar{k}$  but, because we do not observe differences in  $\bar{A}$ ,...

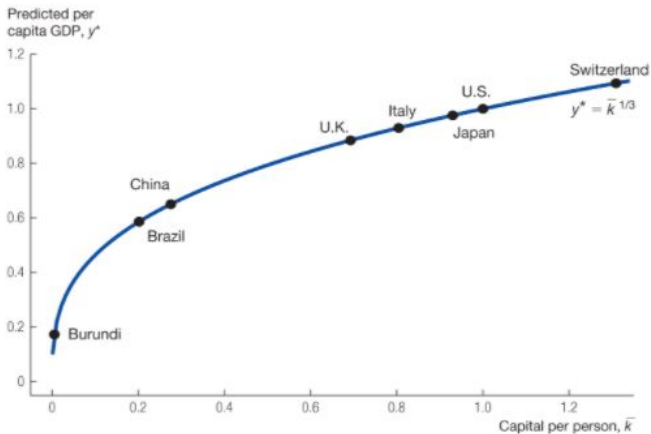
► we will assume that it is the same for all countries.

- In the following exercise, we will calculate the output of country  $c$  relative to the  $US$ :

$$\frac{y_c}{y_{US}} = \left( \frac{\bar{k}_c}{\bar{k}_{US}} \right)^{\frac{1}{3}}$$

# Development Accounting

## Predicted Per Capita GDP in the Production Model



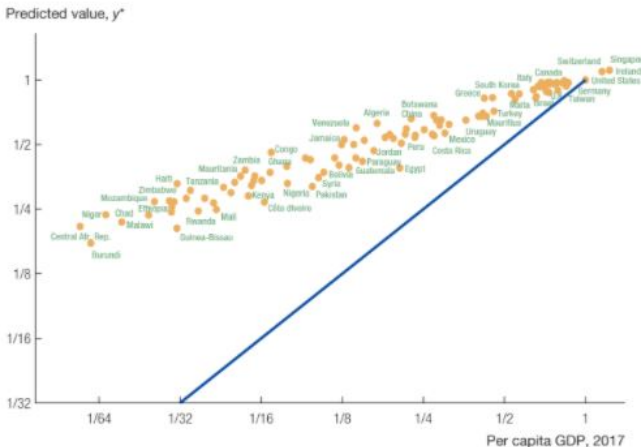
# Development Accounting

The Model's Prediction for Per Capita GDP (U.S. = 1)

| Country      | Observed capital per person, $\bar{k}$ | Predicted per capita GDP $y = \bar{k}^{1/3}$ | Observed per capita GDP |
|--------------|----------------------------------------|----------------------------------------------|-------------------------|
| U.S.         | 1.000                                  | 1.000                                        | 1.000                   |
| Switzerland  | 1.309                                  | 1.094                                        | 1.151                   |
| Japan        | 0.930                                  | 0.976                                        | 0.734                   |
| U.K.         | 0.693                                  | 0.885                                        | 0.714                   |
| Italy        | 0.804                                  | 0.930                                        | 0.680                   |
| Spain        | 0.731                                  | 0.901                                        | 0.640                   |
| China        | 0.276                                  | 0.651                                        | 0.279                   |
| Brazil       | 0.202                                  | 0.587                                        | 0.252                   |
| South Africa | 0.174                                  | 0.559                                        | 0.214                   |
| India        | 0.081                                  | 0.433                                        | 0.117                   |
| Burundi      | 0.005                                  | 0.173                                        | 0.015                   |

# Development Accounting

## The Model's Prediction for Per Capita GDP (U.S. = 1)



## Model with only $k$ and constant $\bar{A}$

- The model without  $\bar{A}$  differences underestimate differences in output per capita.
- On average,  $y$  in industrialized countries is 10 times larger than in non-industrialized countries:

$$10 = \frac{\bar{A}^{ind} (k^{ind})^{\frac{1}{3}}}{\bar{A}^{nind} (k^{nind})^{\frac{1}{3}}}$$

- ▶ If we assume that  $\bar{A}^{ind} = \bar{A}^{nind}$ , then ...

$$10^3 = 1000 = \frac{k^{ind}}{k^{nind}}$$

- ▶ There is no empirical evidence of such large differences.
- Presumably, differences in  $\bar{A}$  play a big role in accounting for differences in  $y$ .
  - ▶ We cannot be certain of this because differences in  $\bar{A}$  are unobserved in general.